

Supplementary Material for FabGAN-ID: Learning the Fabrication Process for Yield-Aware Inverse Design of Multilayer Photonic Sensor Filters

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This supplement reports implementation details and the full result tables behind the main article. Every number is computed on the released benchmark by the accompanying code; the study is fully simulation-based, with the ground-truth process held out from all learning (Sec. S-II). Table references in the main text (Tables S-I–S-XVII) follow the numbering used here. Sections S-XI–S-XVI and Tables S-XII–S-XVII are new in this revision.

S-I. Implementation Details

Solver. Normal-incidence transfer-matrix method (TMM) over 161 wavelengths (400–800 nm), complex-128 precision, JAX `lax.scan` over layers; batched with `vmap`. Layer dispersion: Si₃N₄ Sellmeier (Luke 2015) scaled to the prescribed $n(\lambda_0=550\text{ nm})$; substrate: fused-silica Sellmeier (Malitson 1965); both from `refractiveindex.info` (CC0).

FabGAN. Generator: multilayer perceptron (MLP) $(z_{32}, c_{40}) \rightarrow 128 \rightarrow 128 \rightarrow 40$, leaky-ReLU, 0.30 tanh output head; critic: MLP $(x_{40}, c_{40}) \rightarrow 128 \rightarrow 128 \rightarrow 1$. Conditional Wasserstein GAN with gradient penalty (WGAN-GP), $\lambda_{gp} = 10$ (evaluated at real/generated interpolates sharing the recipe), $n_{critic} = 3$ critic updates per generator update, Adam (10^{-4} , $\beta_1=0.5$, $\beta_2=0.9$), batch 128, 3,000 generator steps. Pooled first/second-moment matching weight $\lambda_{mm} = 5$. Tail calibration (FabGAN-TC): weight 20, applied every 4th generator step after a 25% warmup, calibration bank of 24 recipes $\times$ 16 latent samples, matched $\approx 6\%$ lower-tail fractions of performance *degradation* $\Delta J = J(\hat{\theta}) - J(\theta)$.

Density baselines. Diagonal and full-covariance Gaussians (mean and covariance fitted to the same 400 error vectors; full covariance regularized by $10^{-9}I$, sampled by Cholesky factor); a five-component full-covariance Gaussian mixture (expectation-maximization, EM; `reg_covar` = 10^{-5}); and a nonparametric bootstrap that resamples the 400 observed error vectors with replacement. All include the fitted systematic mean.

Robustification. Pathwise CVaR_{5%} ascent: $K = 256$ reparameterized draws per step (a *fresh* latent minibatch each step, so the tail set is re-sampled, not frozen), $\lceil 0.05K \rceil = 13$ tail samples averaged, projected Adam (2×10^{-3}), 250 steps from the nominal optimum. Mean-variance ablation: $\mu_K - \sigma_K$.

APG policy. Node features $(t_i^*, n_i^*, \lambda_c)$ over the 20-node layer chain; two graph-convolutional-network (GCN) layers (width 64, normalized path adjacency with self-loops), mean pooling, MLP head (width 128) with tanh output scaled to 15% of the design box. Adam (10^{-3}), 400 steps, $K = 128$ pathwise draws per specification per step. SAC baseline: twin Q-critics with the same chain-GCN encoder, automatic entropy tuning, replay 2×10^4 , batch 128, 4,200 environment steps, episode length 6, $K = 48$ -draw CVaR reward. The two learners are matched on *wall-clock*, not on per-update sample count or random seed (Sec. S-VII).

Surrogate (protocol study). Random Fourier features (64 frequencies) over (c_{40}, λ) , tanh MLP 256×256 , amplitude head parameterized multiplicatively around the empty-stack solution with the substrate boundary hard-encoded; recursion residual at label-free collocation designs sampled uniformly from the design box (weight 10^{-2}); Adam (10^{-3}), 2,000 steps, batch 16 designs $\times$ 24 wavelengths.

S-II. Hidden True-Process Parameters

Systematic: thickness rate bias +2%; index drift slope -0.04 across the stack; intermixing fraction 0.12 . Random: first-order autoregressive (AR(1)) thickness noise $\rho = 0.6$, $\sigma = 2\%$ scaled by $\sqrt{t_i}/60$ nm; index noise centered log-normal $\sigma = 0.02$; particulate events with probability 5% per run adding $\mathcal{N}(+8\%, 2\%)$ thickness error to one random layer. These magnitudes are consistent with the systematic and random deposition-error scales reported for the variable-index SiN_x (HDPCVD) platform of Yesilyurt *et al.* (Nanophotonics 2023) and the in-situ monitoring literature (Wilbrandt *et al.*, Appl. Opt. 2008). The joint distribution is non-Gaussian, non-independent, and design-dependent, and is withheld from all learners; evaluation always draws fresh corruptions from it.

S-III. Released Benchmark: File-Level Documentation

data/benchmark_designs.npz, arrays **d_um** (300×20): layer thicknesses in μm ; **n0** (300×20): layer indices at $\lambda_0 = 550$ nm; **T** (300×161): exact transmittance spectra (300 spectra, i.e., 48,300 wavelength-transmittance samples); **lam_um** (161): wavelength grid in μm . A CSV export (**benchmark_designs.csv**) accompanies the archive. **data/process_traces.npz**, arrays **recipe_d**, **recipe_n**, **fab_d**, **fab_n** (each 400×20): the synthetic historical traces, i.e., recipes and their fabricated realizations drawn from the withheld ground-truth simulator. **Provenance**: material dispersion derives from the public-domain refractiveindex.info database (Si₃N₄: Luke 2015; SiO₂: Malitson 1965); design ranges follow the variable-index SiN_x platform of Yesilyurt *et al.* (Nanophotonics 2023); all numerical data are generated by the authors’ simulation framework, no experimental wafer data are used or implied.

S-IV. Computational Cost

All experiments, solver validation, twin training, all density baselines, robustification, the oracle and mean-only controls, policy learning, and the surrogate protocol study, complete in approximately 1 CPU-hour in total on a two-core commodity processor with no accelerator; all randomness is seeded.

S-V. Detailed Result Tables

Tables S-I–S-XI give the full detail behind the main text, in the order referenced there.

Table S-I: Twin Tail Fidelity vs. Trace Budget M (2,000-Step GANs, 15 Held-Out Designs $\times$ 400 True Draws). Physics-in-the-loop Tail Calibration Helps Only at $M = 50$. Lower Is Better.

M	Twin	$W_1(J)$	$ \Delta P_5 $	$ \Delta \text{CVaR}_{5\%} $
50	FabGAN	0.0224	0.0182	0.0181
50	FabGAN-TC	0.0189	0.0183	0.0176
100	FabGAN	0.0144	0.0144	0.0148
100	FabGAN-TC	0.0206	0.0188	0.0177
400	FabGAN	0.0137	0.0152	0.0161
400	FabGAN-TC	0.0200	0.0163	0.0177

Table S-II: Process-Twin Fidelity Against the Hidden True Process (30 Held-Out Designs $\times$ 400 Draws, Identical Protocol; All Models Fitted to the Same 400 Traces). *Corr. err* Is the Frobenius Relative Error of the Error-Vector Correlation Matrix. Lower Is Better; Best in Bold.

Density model	Corr. err	$W_1(J)$	$ \Delta P_5 $	$ \Delta \text{CVaR}_{5\%} $
Gaussian (diagonal)	0.622	0.0186	0.0194	0.0202
Gaussian (full covariance)	0.347	0.0179	0.0187	0.0204
Gaussian mixture (5 comp.)	0.481	0.0181	0.0184	0.0199
Nonparametric bootstrap	0.352	0.0175	0.0194	0.0203
FabGAN (ours)	0.708	0.0154	0.0122	0.0123

Table S-III: True-Process Yield of Robustified Designs (532 nm, 2,000 Common Draws From the Hidden Process). Higher Is Better. FabGAN-TC Is Included as an Honest Ablation and Is *Worse* Than the Plain FabGAN Twin at This Trace Budget ($M = 400$), Consistent With Table S-I.

Design / twin	$\mu \uparrow$	σ/μ [%] $\downarrow$	$P_5 \uparrow$	$\text{CVaR}_{5\%} \uparrow$
Nominal optimum θ^*	0.9048	1.8	0.8742	0.8571
Gaussian (diag) twin, CVaR	0.9329	1.1	0.9142	0.9050
Gaussian (diag) twin, $\mu - \sigma$	0.9336	1.2	0.9124	0.9024
Gaussian (full) twin, CVaR	0.9347	1.1	0.9164	0.9091
FabGAN-TC twin, CVaR	0.9355	1.3	0.9131	0.9040
FabGAN twin, $\mu - \sigma$	0.9429	0.9	0.9272	0.9210
FabGAN twin, CVaR (ours)	0.9405	0.9	0.9251	0.9187

Table S-IV: Paired Bootstrap 95% Confidence Intervals for True-Process $\text{CVaR}_{5\%}$ Differences (2,000 Common Draws, $B = 4,000$ Resamples). All Differences Are Significant. The Mean-Variance Objective Is Marginally but Significantly Ahead of CVaR, Roughly Sixfold Smaller Than the Twin Effect (FabGAN – Gaussian).

Contrast	$\Delta \text{CVaR}_{5\%}$	95% CI
FabGAN – nominal	+0.0615	[+0.0579, +0.0651]
FabGAN – Gaussian (diag)	+0.0136	[+0.0115, +0.0160]
FabGAN – Gaussian (full)	+0.0095	[+0.0080, +0.0112]
FabGAN $\mu - \sigma$ – FabGAN CVaR	+0.0024	[+0.0019, +0.0028]

Table S-V: True-Process Oracle and Bias-Correction Controls (8,000 Common Draws From the Hidden Process; Larger Reference Than Table S-III, Hence Slightly Different Absolute Levels). *Regret* Is the Oracle $\text{CVaR}_{5\%}$ Minus the Design's. The Oracle Is a Derivative-Free CVaR Optimizer With Full Access to the Hidden Simulator.

Design	$\text{CVaR}_{5\%} \uparrow$	Regret $\downarrow$
Nominal optimum	0.8566	0.060
Mean-only bias correction	0.8679	0.049
Gaussian (diag) twin, CVaR	0.9025	0.014
Gaussian (full) twin, CVaR	0.9072	0.009
FabGAN twin, CVaR (ours)	0.9165	0.000
True-process oracle	0.9165	,

Table S-VI: Device-Level Sensor Metrics and Hard Pass/Fail Filter Yield Under the Hidden Process (5,000 True-Process Draws). Pass Criterion: Worst-Case Stop-Band Leakage $\leq 9\%$ *and* Mean Pass-Band Transmission $\geq 90\%$ on Each Fabricated Draw. Nominal-Design Spectral Values Are Deterministic; Yield Is Over the Ensemble.

Design	$T(532\text{ nm})$	Max stop-band leakage	Mean pass-band T	Pass/fail yield
Nominal optimum	0.041	0.063	0.958	0.16%
FabGAN-robustified (ours)	0.023	0.039	0.921	63.7%

Table S-VII: Multi-Specification Robustification: True-Process $\text{CVaR}_{5\%}$ of the Nominal Optimum vs. the FabGAN-Robustified Design, Optimized Independently at Four Specifications. Gains Are Positive at Every Specification.

Spec.	Nominal	FabGAN (ours)	Gain
520 nm	0.8259	0.9156	+10.9%
532 nm	0.8571	0.9187	+7.2%
580 nm	0.7799	0.8726	+11.9%
620 nm	0.7716	0.8652	+12.1%

Table S-VIII: Risk-Level Sweep: True-Process CVaR_{α} (532 nm, 2,000 Common Draws) at Several Tail Levels. Robustification Used $\alpha = 5\%$; the FabGAN Design Dominates the Best Gaussian Twin at *Every* Level, so the Advantage Is Not an Artifact of Aligning α With the Objective.

Design	$\alpha=1\%$	2.5%	5%	10%	20%
Nominal optimum	0.8321	0.8461	0.8571	0.8689	0.8796
Gaussian (full) twin	0.8943	0.9036	0.9091	0.9145	0.9199
FabGAN twin (ours)	0.9079	0.9144	0.9187	0.9229	0.9270

Table S-IX: Latent-Dimension Ablation: Twin Tail Fidelity vs. Latent Width d_z (1,500-Step Plain WGAN Trainings, 12 Held-Out Designs $\times$ 400 True Draws). Fidelity Degrades Only at $d_z=16$ and Is Statistically Flat for $d_z \geq 32$ (Across and Beyond the 40-Dimensional Design Space), so $d_z=32$ Is Not a Bottleneck. Lower Is Better.

	$d_z=16$	$d_z=32$	$d_z=48$	$d_z=64$
$ \Delta P_5 $	0.0259	0.0174	0.0148	0.0161
$ \Delta \text{CVaR}_{5\%} $	0.0252	0.0191	0.0167	0.0201

Table S-X: Zero-Shot Recovery of the Attainable True-Process $\text{CVaR}_{5\%}$ Gain at Held-Out Specifications [%]. Per-Specification Re-Optimization = 100%. Interpolation (580 nm) Transfers; Basin Changes (520, 620 nm) Do Not.

Spec.	Static	SAC	APG (5 specs)	APG (11 specs)
520 nm	3.15	-560.06	-115.39	-137.92
580 nm	31.32	-360.89	65.23	86.75
620 nm	-95.81	-450.90	-121.30	-35.69

Table S-XI: Capability Comparison Against the State of the Art. ✓ = Provided; ~ = Partial; ✗ = Absent. This Is the Qualitative Counterpart to Table I of the Main Article.

Framework	Process model learned from data	Non-Gauss., correlated, design-cond.	Corruption operator differentiable	Yield-tail (CVaR) objective	Exact-gradient physics in loop	Policy learning through loop	Validated vs. held-out true process	Open benchmark released
Classical needle/gradual synth. [Macleod 2010]	✗	✗	✗	✗	✓	✗	✗	✗
Error-retrained NN film des. [Yesilyurt 2023]	✗	✗	✗	✗	✓	✗	✗	✗
Foundry-constrained robust ID [Schubert 2022]	✗	~	✗	✗	✓	✗	✗	✗
CNN fab-variation predictor [Gostimirovic 2022]	✓	~	~	✗	✗	✗	~	✗
Fabrication-aware diffusion design [Seo 2026]	✓	✓	~	✗	~	✗	✗	✗
Gen-Fab cGAN variation twin [Azimi 2026]	✓	✓	✗	✗	✗	✗	~	✗
Model-free RL photonic design [Hooten 2021; Li 2023]	✗	✗	✗	✗	~	~	✗	✗
Data-centric sensor ID [Mahim 2025]	✗	✗	✗	✗	✗	✗	✗	✓
FabGAN-ID (this work)	✓	✓	✓	✓	✓	✓	✓	✓

S-VI. Trace-Scarcity and Latent-Dimension Ablations

Table S-I evaluates the physics-in-the-loop tail-calibration objective across historical-trace budgets $M \in \{50, 100, 400\}$ (15 held-out designs $\times$ 400 true-process draws per condition; Fig. S-1). Calibration improves induced-performance fidelity only in the scarce-trace regime ($M = 50$); with abundant traces the purely adversarial twin is already tail-faithful, and Table S-III confirms that at $M = 400$ the tail-calibrated variant is in fact *worse* than the plain FabGAN twin on true-process yield. Table S-IX sweeps the latent width d_z : tail fidelity degrades only at $d_z=16$ (below the 40-dimensional design/error space) and is statistically flat for $d_z \geq 32$, so the reported $d_z=32$ sits above the bottleneck.

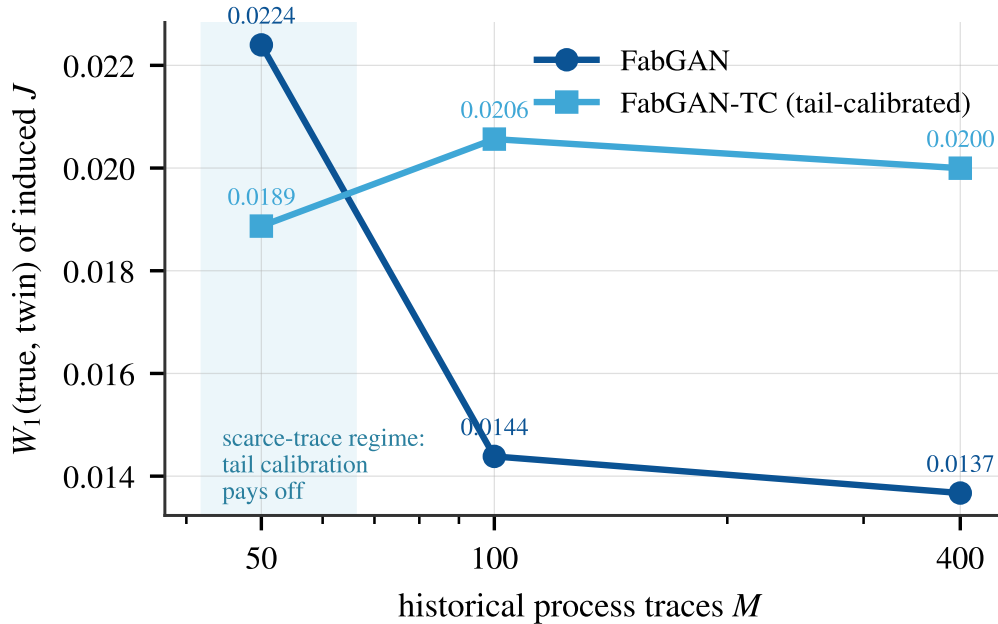


Figure S-1: Trace-scarcity study: induced-performance fidelity of the twin vs. number of historical traces M , with and without physics-in-the-loop tail calibration. The calibration pays off only in the scarce-trace regime ($M = 50$).

S-VII. Correction-Policy Training and Transfer Detail

Fig. S-2 shows policy training in the FabGAN loop, analytic policy gradients converge ($0.77 \rightarrow 0.88$ mean CVaR reward) where model-free SAC fails to improve ($0.73 \rightarrow 0.64$) at comparable wall-clock budget, with per-specification zero-shot recovery of the attainable true-process CVaR_{5%} gain in Table S-X. As noted

in Sec. S-I, the comparison controls wall-clock, not per-update sample count or seed; a fully sample- and seed-matched study is future work. Interpolation (580 nm) transfers; basin changes (520, 620 nm) do not, motivating cheap per-specification re-optimization in the differentiable regime.

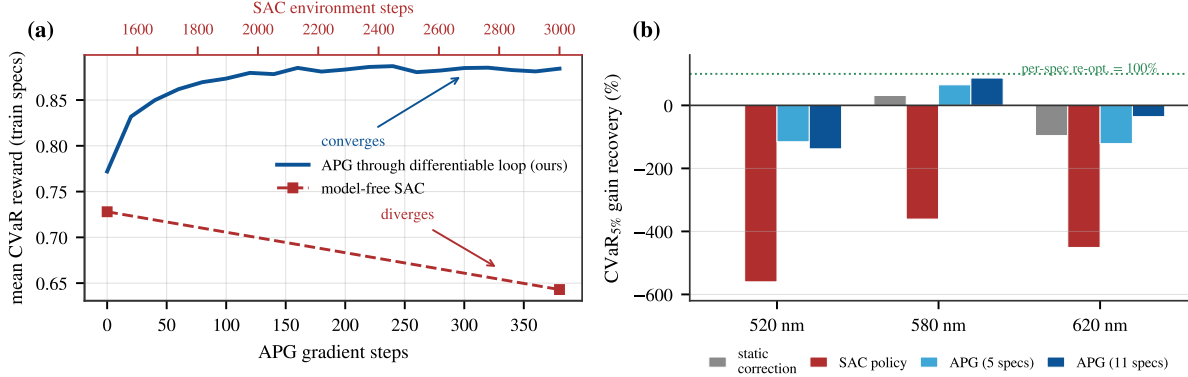


Figure S-2: a) Policy training: analytic policy gradients vs. model-free SAC; b) zero-shot transfer recovery at held-out specifications.

S-VIII. Surrogate Protocol Study

Fig. S-3 contrasts held-out R^2 under row splitting vs. strict held-out-design splitting for the data-only and recursion-informed surrogates. Row splitting overstates utility catastrophically; the exact differentiable solver (used by FabGAN-ID) is both exact and $18\times$ faster. This conclusion is scoped to the tested regime (240 labeled designs, 40 dimensions, a fast differentiable solver) and does not claim that surrogates are unhelpful for expensive, non-differentiable, or data-abundant forward models.

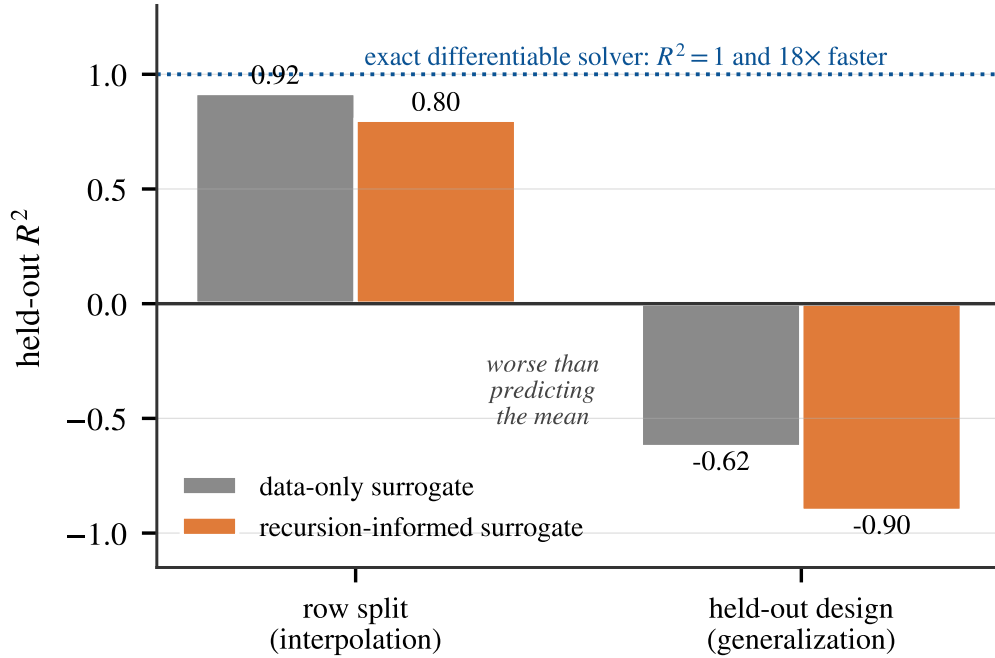


Figure S-3: Surrogate protocol study: held-out R^2 under row splitting vs. strict held-out-design splitting, for data-only and recursion-informed surrogates.

S-IX. Design-Gradient Validation

Because robustification differentiates through the learned twin (not only the physics), the twin’s *design* gradient must be correct, not merely its samples. We compare directional derivatives of the mean merit taken through the FabGAN twin (reverse-mode autodiff) against common-random-number central finite differences of the withheld true simulator, at 14 held-out recipes along random unit directions in the 40-dimensional design space. The twin and true directional derivatives correlate at Pearson $r = 0.91$, with 79% sign agreement; residual magnitude scatter reflects the finite-sample noise of the true-process finite differences. The learned $\partial\tilde{\theta}/\partial\theta$ therefore supplies usable design gradients, validating the pathwise robustification of the main text.

S-X. Nominal Inverse-Design Convergence

The probe-seeded adjoint engine underlies every downstream result: it supplies the nominal optimum θ^* that is subsequently robustified. Fig. S-4 shows its convergence at the 532-nm specification against an equal-budget random search (the quantitative counterpart is Table I(C) of the main article). Two hundred random probes locate the promising basin; four gradient refinements driven by the exact adjoint of the transfer-matrix recursion then exploit it, reaching $J^* = 0.956$ in 360 solver queries, whereas random search plateaus at 0.824 under the same budget and does not match J^* within 30,000 queries ($> 83\times$ query-efficiency advantage). This confirms that global probing discovers the basin while exact gradients, not stochastic search, are what exploit it.

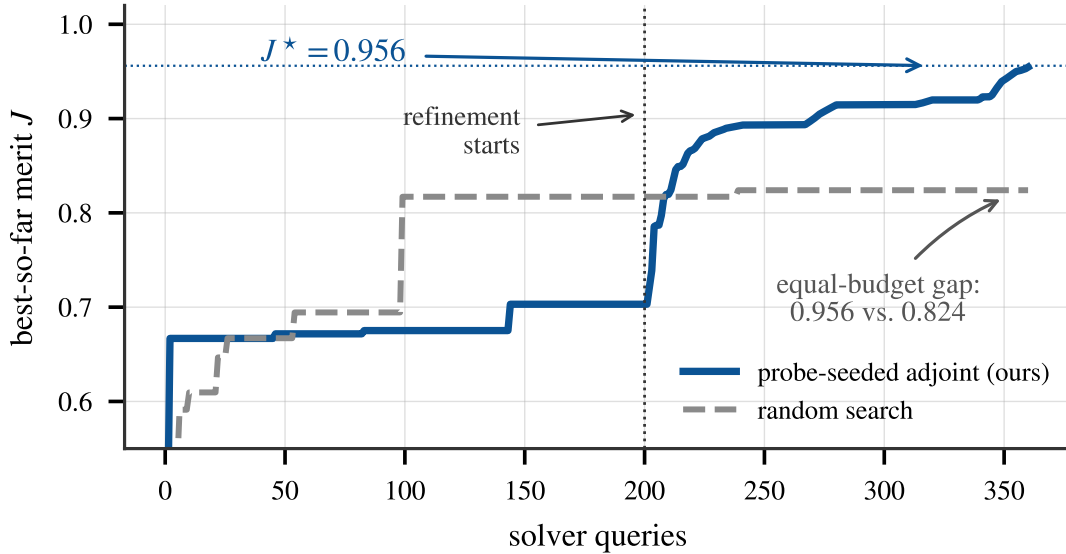


Figure S-4: Nominal inverse design at 532 nm: best-so-far merit J versus solver queries for the probe-seeded adjoint engine (solid) and an equal-budget random search (dashed); the dotted vertical line marks the start of gradient refinement, and the dotted horizontal line the attained optimum $J^* = 0.956$. The optimized transmittance spectrum is shown in Fig. 1(c) of the main article.

S-XI. Finite- K CVaR Gradient Study

The robustification gradient is the exact reverse-mode derivative of the *empirical* CVaR, the average of the $[\alpha K]$ smallest of K reparameterized merit draws. This function of the design is piecewise smooth (differentiable wherever the order statistics are untied), so the estimator is well defined almost everywhere; however, the finite- K tail selection makes it a biased estimator of the population CVaR gradient, with bias vanishing as K grows (Hong and Liu, *Manage. Sci.*, 2009). We therefore quantify its finite- K behavior

empirically at the nominal 532-nm design: for each $K \in \{64, 256, 2048\}$, $R = 100$ independent gradient estimates are drawn from disjoint latent streams, and all are compared against a reference direction defined as the mean of the 100 $K=2048$ estimates. Table S-XII reports (i) the relative ℓ_2 deviation of the mean estimate from the reference (finite- K bias; zero for $K=2048$ by construction), (ii) the mean cosine similarity of individual estimates to the reference, and (iii) the per-estimate noise-to-signal ratio. At the operating point $K = 256$ the mean estimate deviates from the large- K reference by 1.3% in norm with mean cosine similarity 0.98: the estimator is low-bias and direction-faithful in practice, which is what projected stochastic ascent requires; the revised main text accordingly avoids any claim of exact unbiasedness.

Table S-XII: Finite- K Behavior of the Pathwise CVaR_{5%} Gradient Estimator at the Nominal 532-nm Design ($R = 100$ Repetitions per K ; Reference = Mean of the $K=2048$ Estimates).

K	Rel. bias of mean ↓	Mean cosine to ref. ↑	Noise-to-signal ↓
64	0.193	0.894	0.479
256 (operating point)	0.013	0.984	0.273
2048	0 (by constr.)	0.998	0.108

S-XII. Extended Density Baselines

Four additional density models are fitted to the same 400 traces and scored under the identical protocol of Table S-II (30 held-out designs $\times$ 400 fresh true-process draws), with every hyperparameter chosen from the traces alone: a Gaussian copula and a t copula ($\nu = 5$) combining empirical marginals with a rank-based correlation matrix; a kernel density estimate over the pooled error vectors with Silverman’s bandwidth; and a RealNVP normalizing flow (Dinh *et al.*, ICLR 2017; six affine coupling layers of hidden width 96 on standardized data, Adam, 2,500 steps). Table S-XIII shows that all four remain less accurate than the learned twin on every tail metric, so the main-text tail-error reduction holds against the strongest of eight density models (the five-component Gaussian mixture of Table I(A) remains the strongest baseline overall).

Table S-XIII: Extended Density Baselines Under the Protocol of Table S-II (30 Held-Out Designs $\times$ 400 True Draws; Mean Absolute Estimation Error, Lower Is Better). The Gaussian-Mixture Row Is Repeated From Table I(A) for Reference.

Corruption model	$ \Delta P_5 $	$ \Delta \text{CVaR}_{5\%} $	$W_1(J)$
Gaussian copula (empirical marginals)	0.0191	0.0203	0.0177
t copula ($\nu=5$, empirical marginals)	0.0187	0.0203	0.0181
Kernel density (Silverman)	0.0260	0.0292	0.0194
RealNVP flow	0.0197	0.0200	0.0186
Gaussian mixture (5 comp.)	0.0184	0.0199	0.0181
FabGAN twin	0.0122	0.0123	0.0154

S-XIII. Statistical Variability: Training Seeds, Trace Datasets, and Learning Curve

Tail fidelity is re-measured under replication at a fixed, reduced training budget (1,500 generator steps; 12 held-out designs $\times$ 400 fresh true-process draws per design, with common reference statistics). Because this protocol differs from that of Table S-II (3,000 steps, 30 designs), the parametric baselines are re-scored under the *identical* protocol for a like-for-like comparison. Table S-XIV shows: (a) across five GAN training seeds, mean $|\Delta P_5| = 0.0143 \pm 0.0021$ and $|\Delta \text{CVaR}_{5\%}| = 0.0154 \pm 0.0033$ (mean $\pm$ sd), with every individual seed more accurate than the strongest parametric baseline on both metrics; (b) the same holds across three independently regenerated 400-trace datasets; (c) the trace-budget learning curve (three seeds per budget) improves steeply from $M=50$ to $M=100$ and is approximately flat beyond $M \approx 200$, locating the returns to

additional traces. Full end-to-end (robustification-level) replication over many seeds is noted as future work in the main text.

Table S-XIV: Replication of Twin Tail Fidelity at a Fixed 1,500-Step Budget (12 Held-Out Designs $\times$ 400 True Draws; Mean Absolute Estimation Error, Lower Is Better). Baselines Re-Scored Under the Identical Protocol.

Study	Instance	$ \Delta P_5 $	$ \Delta \text{CVaR}_{5\%} $
(a) GAN training seed	seed 7	0.0174	0.0191
	seed 17	0.0131	0.0124
	seed 27	0.0149	0.0150
	seed 37	0.0119	0.0119
	seed 47	0.0140	0.0186
(b) Regenerated trace dataset	dataset 1	0.0136	0.0164
	dataset 2	0.0162	0.0188
	dataset 3	0.0153	0.0181
(c) Learning curve (mean of 3 seeds)	$M = 50$	0.0227	0.0235
	$M = 100$	0.0155	0.0160
	$M = 200$	0.0141	0.0152
	$M = 400$	0.0151	0.0155
Baselines (same protocol)	Full-covariance Gaussian	0.0192	0.0234
	Gaussian mixture (5 comp.)	0.0182	0.0216

S-XIV. Process-Family Robustness Sweep

To test whether the conclusions are tied to the six scripted error mechanisms, the entire pipeline (trace regeneration, twin retraining at the full 3,000-step budget, refitting of the full-covariance Gaussian, robustification of the nominal 532-nm optimum under each twin, and evaluation on 2,000 fresh draws from the variant process) is repeated under six hidden-process variants plus an unmodified control: rare-particulate events removed ($p = 0$) or doubled in probability ($p = 10\%$) or severity (+16% mean), thickness-noise correlation lowered ($\rho = 0.3$) or raised ($\rho = 0.85$), and index-noise scale doubled ($\sigma = 0.04$). Table S-XV reports true-process tail statistics. Three observations: every robustification improves markedly on the nominal design under every process; the FabGAN-robustified design attains the higher P_5 and $\text{CVaR}_{5\%}$ in six of the seven processes, including the control (which replicates the main result on a regenerated trace set) and, notably, the no-rare-event process; and under the doubled index-noise variant the Gaussian-twin design is marginally ahead ($\text{CVaR}_{5\%}$ 0.874 vs. 0.868), so the learned-twin advantage is robust but not universal. All variants remain members of one simulator family, a scope limitation stated in the main text.

Table S-XV: Process-Family Sweep: True-Process Tail Statistics After Identical Pathwise $\text{CVaR}_{5\%}$ Robustification Under Each Retrained Twin (2,000 Fresh Draws per Design and Process; Higher Is Better; Better Twin in Bold).

Process variant	Change	Nominal		Robust vs. Gaussian		Robust vs. FabGAN	
		P_5	$\text{CVaR}_{5\%}$	P_5	$\text{CVaR}_{5\%}$	P_5	$\text{CVaR}_{5\%}$
Control (unmodified)	—	0.874	0.857	0.915	0.908	0.923	0.915
No rare events	$p = 0$	0.875	0.859	0.915	0.906	0.919	0.913
Rare events 2 $\times$	$p = 10\%$	0.873	0.856	0.917	0.910	0.923	0.917
Severity 2 $\times$	$\mu_{fl} = +16\%$	0.872	0.853	0.916	0.908	0.919	0.910
Correlation low	$\rho = 0.3$	0.885	0.875	0.918	0.911	0.928	0.923
Correlation high	$\rho = 0.85$	0.844	0.809	0.907	0.889	0.914	0.898
Index noise 2 $\times$	$\sigma = 0.04$	0.835	0.807	0.889	0.874	0.880	0.868

S-XV. Yield Counts, Confidence Intervals, and Threshold Sensitivity

Table S-XVI restates the simulated pass/fail yields of the main text as raw counts with Wilson 95% confidence intervals and sweeps both thresholds. At the specification of the main text (stop-band leakage $\tau_s = 0.09\%$, mean pass-band transmission $\tau_p = 90\%$) the nominal optimum passes 8/5,000 draws (0.16%, CI [0.08, 0.32]%) and the FabGAN-robustified design 3,186/5,000 (63.7%, CI [62.4, 65.0]%). Across the full 4×3 grid the FabGAN-robustified design attains the strictly higher simulated yield at all twelve threshold pairs, so the ordering is not an artifact of the chosen thresholds; the absolute yield values, as expected, vary strongly with the thresholds.

Table S-XVI: Simulated Pass/Fail Yield vs. Pass/Fail Thresholds (5,000 Paired Draws From the Hidden Process; Wilson 95% CIs). τ_s : Maximum Stop-Band Leakage; τ_p : Minimum Mean Pass-Band Transmission. Main-Text Specification in Bold.

τ_s [%]	τ_p [%]	Nominal optimum		FabGAN-robustified	
		pass	yield [%] (CI)	pass	yield [%] (CI)
0.06	85	0	0.00 (0.00–0.08)	377	7.54 (6.84–8.30)
0.06	90	0	0.00 (0.00–0.08)	373	7.46 (6.76–8.22)
0.06	92	0	0.00 (0.00–0.08)	358	7.16 (6.48–7.91)
0.09	85	8	0.16 (0.08–0.32)	3202	64.0 (62.7–65.4)
0.09	90	8	0.16 (0.08–0.32)	3186	63.7 (62.4–65.0)
0.09	92	5	0.10 (0.04–0.23)	3050	61.0 (59.6–62.3)
0.12	85	462	9.24 (8.47–10.1)	4524	90.5 (89.6–91.3)
0.12	90	455	9.10 (8.33–9.93)	4500	90.0 (89.1–90.8)
0.12	92	360	7.20 (6.52–7.95)	4277	85.5 (84.5–86.5)
0.15	85	1780	35.6 (34.3–36.9)	4848	97.0 (96.5–97.4)
0.15	90	1745	34.9 (33.6–36.2)	4815	96.3 (95.7–96.8)
0.15	92	1377	27.5 (26.3–28.8)	4540	90.8 (90.0–91.6)

S-XVI. Transfer to a Second Design Family: 600-nm Band-Pass

To probe generalization beyond the notch objective, the identical pipeline is applied to a 600-nm band-pass task on the same platform: pass band 585–615 nm, stop bands 400–555 nm and 645–800 nm, merit $J_{bp} = \frac{1}{2}\overline{T}_{\text{pass}} + \frac{1}{2}(1 - \overline{T}_{\text{stop}})$. The probe-seeded adjoint engine attains $J_{bp}^* = 0.835$; the design is then robustified by the same pathwise CVaR_{5%} ascent under the FabGAN twin and under the full-covariance Gaussian twin, and all three designs are scored on 1,500 fresh draws from the hidden process (Table S-XVII). Robustification transfers: the FabGAN twin raises the true-process CVaR_{5%} from 0.701 to 0.751 (+7.2% relative), essentially matching the notch-task gain. On this objective the full-covariance Gaussian twin is statistically comparable (0.753), unlike on the notch task where the FabGAN design dominated; the size of the learned-twin advantage over the best parametric twin is therefore objective-dependent, which the revised main text states explicitly.

Table S-XVII: Band-Pass Objective (600 nm) Under the Hidden Process (1,500 Fresh Draws per Design; Higher Is Better).

Design	μ	P_5	CVaR _{5%}
Nominal optimum ($J_{bp}^* = 0.835$)	0.763	0.720	0.701
Robust vs. full-covariance Gaussian twin	0.798	0.765	0.753
Robust vs. FabGAN twin	0.794	0.764	0.751